

AA-NITROFURANTOIN

Nitrofurantoin Tablets BP 100mg
Urinary Antibacterial

Pharmacology: Nitrofurantoin is bacteriostatic in low concentrations (1:100 000 to 1: 200 000) and in vitro is considered to be bactericidal in higher concentrations. Its presumed mode of action is based on its interference with several bacterial enzyme systems. Nitrofurantoin is effective against most strains of *E. coli*, but many strains of *Enterobacter* and *Klebsiella* and most strains of *Proteus* and *Pseudomonas* are resistant. Antimicrobial activity is highest in acid urine. Microorganisms develop only a limited resistance to furan derivatives.

The efficacy of nitrofurantoin is dependent upon achieving an adequate concentration of the drug in the urine. Therefore urine concentrations and recoveries, rather than blood or serum concentrations, are important indicators of bioavailability.

Nitrofurantoin is rapidly and almost completely absorbed from the gastrointestinal tract. Very low blood concentrations and high urine concentrations have been observed uniformly in animals and man. The serum half life of nitrofurantoin in humans is approximately 20 minutes, and it is rapidly excreted in the urine. Because of the low blood concentrations and short half-life, the major therapeutic effect of nitrofurantoin occurs within the urinary tract. Between 30 and 50% of an orally administered dose of nitrofurantoin can be recovered intact from the urine. Renal excretion involves glomerular filtration and tubular secretion. Urinary recovery of nitrofurantoin is linearly related to creatinine clearance, and little or no drug is excreted by patients with marked renal failure. Nitrofurantoin is degraded by all tissues of the body (except blood) into inactive metabolites which may color the urine brown.

The absorption of nitrofurantoin macrocrystals, a larger crystal form of nitrofurantoin, is slower and the excretion is somewhat less than that of nitrofurantoin when the two are compared. The reduced incidence of gastrointestinal intolerance with nitrofurantoin macrocrystals is probably due to delayed and decreased absorption; this, however, does not significantly reduce clinical effectiveness. When administered orally, the possibility of nausea is diminished if nitrofurantoin is given with meals or if the macrocrystalline form is given.

Indications: The treatment of pyelonephritis, pyelitis and cystitis when due to susceptible organisms. Not indicated for the treatment of associated renal cortical or perinephric abscesses, nor in prostatitis.

Contraindications: Anuria, oliguria, or extensive impairment of renal function (creatinine clearance under 40ml/min.). Treatment of this type of patient carries an increased risk of toxicity because of impaired excretion of drug. For the same reason, the drug is much less effective under these circumstances. Contraindicated at term in pregnant patients as well as in infants under 1 month of age, because of the possibility of hemolytic anemia due to immature enzyme systems (glutathione instability). Known hypersensitivity to nitrofurantoin.

Precautions: Hemolytic anemia of the primaquine sensitivity type has been induced by nitrofurantoin. The hemolysis appears to be linked to a glucose-6-phosphate dehydrogenase deficiency in the affected patients' red blood cells. This deficiency is found in 10% of Blacks and in a small percentage of ethnic groups of Mediterranean and Near-Eastern origin. Any sign of hemolysis is an indication to discontinue the drug. Hemolysis ceases when the drug is withdrawn.

Predisposing conditions such as renal impairment, anemia, diabetes, electrolyte imbalance, vitamin B deficiency, and debilitating disease may enhance the occurrence of peripheral neuropathy.

Peripheral neuropathy may occur with nitrofurantoin therapy; this may become severe or irreversible. A fatality has been reported. If numbness or tingling occurs, discontinue the drug.

Pregnancy: The safety of nitrofurantoin during pregnancy has not been established. It should not be used in women of child bearing potential unless the expected benefits outweigh the possible hazards.

Lactation: Nitrofurantoin is contraindicated in infants with G-6-PD deficiency as hemolytic anemia has been reported.

Drug Interactions: Do not administer nitrofurantoin concomitantly with drugs which may produce impaired renal function. Probenecid or sulphinpyrazone may reduce the excretion of nitrofurantoin and should not be given concomitantly. Magnesium trisilicate may reduce the absorption of nitrofurantoin.

Nitrofurantoin and the quinolone antibiotics are antagonistic in vitro and should not be used together.

Adverse Effects: Gastrointestinal: Anorexia, nausea, emesis are the most frequent reactions; less frequently, abdominal pain and diarrhea; rarely, hepatitis. This dose-related toxicity reaction can be minimized by reduction of dosage, especially in the female patient. Hypersensitivity: pulmonary sensitivity reactions, which can be acute, subacute, or chronic. Acute reaction is commonly manifested by fever, chills, cough, chest pain, dyspnea, pulmonary infiltration with consolidation or pleural effusion on x-ray, and eosinophilia. The acute reactions usually occur within the first week of treatment and resolve with cessation of the drug therapy. Subacute or chronic pulmonary reaction is associated with prolonged therapy. Insidious onset of malaise, dyspnea on exertion, cough, altered pulmonary function, and roentgenographic and histologic findings of diffuse interstitial pneumonitis or fibrosis or both are common manifestations. Impaired pulmonary function may result even after cessation of the drug therapy. Dermatologic: maculopapular, erythematous, or eczematous eruption, pruritus, urticaria, angioedema. Other sensitivity reactions: anaphylaxis, asthmatic attack in patients with history of asthma, cholestatic jaundice, drug fever, arthralgia. Hematologic: hemolytic anemia, granulocytopenia, eosinophilia, megaloblastic anemia. Return of the blood picture to normal has followed cessation of therapy. Neurological: peripheral neuropathy, headache, dizziness, nystagmus, and drowsiness. Miscellaneous: transient alopecia. As with other antimicrobial agents, superinfections by resistant organisms may occur. With nitrofurantoin, however, these are limited to the genitourinary tract because suppression of normal bacterial flora elsewhere in the body does not occur. *Pseudomonas* is the organism most commonly implicated in superinfections in patients treated with nitrofurantoin.

Overdose: Symptoms: Occasional incidents of acute overdosage of nitrofurantoin have not resulted in any specific symptomatology other than vomiting.

Treatment: In case vomiting does not occur soon after an excessive dose, induction of emesis is recommended. There is no specific antidote for

nitrofurantoin but a high fluid intake should be maintained to promote urinary excretion of the drug, but only in case of overdosage.

Dosage: To minimize gastric upset, administer the drug with food or milk. Adults: 50 to 100mg 4 times a day. Children: 5 to 7mg/kg/24 hours given in divided doses 4 times a day (contraindicated in infants under 1 month).

Continue therapy for at least 1 week and for at least 3 days after sterility of the urine is obtained. Continued infection indicates the need for reevaluation. If the drug is to be used for long-term suppressive therapy, consider a dosage reduction.

Mode of administration: Oral

Supplied: Each round, biconvex, yellow tablets, scored on one side, other side plain contains nitrofurantoin 100mg. Available in HDPE bottles of 28 tablets and 100 tablets. Not all pack sizes are marketed.

Storage Conditions: Store at temperature not exceeding 25°C.

Manufacturer

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